

understanding of the types of bacteria within and around roots could potentially have profound agricultural benefits.

Bacteria in the Nitrogen Cycle

Because nitrogen is required in large amounts for synthesizing proteins and nucleic acids, no mineral deficiency is more limiting to plant growth than a lack of nitrogen. The forms of nitrogen that plants can use include NO_3^- and NH_4^+ . Some soil nitrogen derives from the weathering of rocks, and lightning produces small amounts of NO_3^- that get carried to the soil in rain. However, most of the nitrogen available to plants comes from the activity of bacteria (**Figure 37.11**). This activity is part of the **nitrogen cycle**, a series of natural processes by which certain nitrogen-containing substances from the air and soil are made available to living things, are used by them, and are returned to the air and soil (see Figure 55.14).

Plants commonly acquire nitrogen in the form of NO_3^- (nitrate). Soil NO_3^- is largely formed by a two-step process called *nitrification*, which consists of the oxidation of ammonia (NH_3) to nitrite (NO_2^-), followed by oxidation of NO_2^- to NO_3^- . Different types of *nitrifying bacteria* mediate each step, as shown at the bottom of Figure 37.11. After the roots absorb NO_3^- , a plant enzyme reduces it back to NH_4^+ , which other enzymes incorporate into amino acids and other organic compounds. Most plant species transport nitrogen from roots to shoots via the xylem as NO_3^- or as organic compounds synthesized in the roots. Some soil nitrogen is

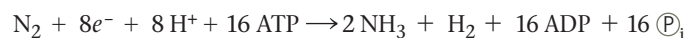
lost, particularly in anaerobic soils, when denitrifying bacteria convert NO_3^- to N_2 , which diffuses into the atmosphere.

In addition to NO_3^- , plants can acquire nitrogen in the form of NH_4^+ (ammonium) through two processes, as shown on the left in Figure 37.11. In one process, *nitrogen-fixing bacteria* convert gaseous nitrogen (N_2) to NH_3 , which then picks up another H^+ in the soil solution, forming NH_4^+ . In the other process, called *ammonification*, decomposers convert the organic nitrogen from dead organic material into NH_4^+ .

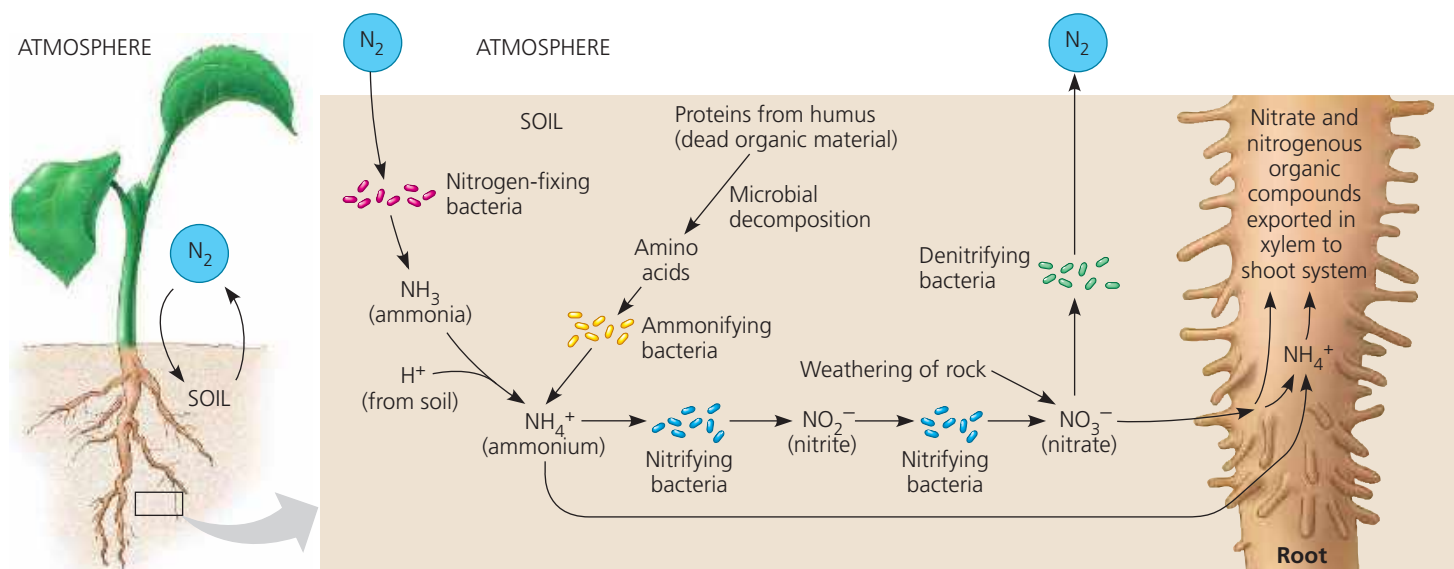
Bacteria and Nitrogen Fixation

Although Earth's atmosphere is 79% nitrogen, plants cannot use free gaseous nitrogen (N_2) because there is a triple bond between the two nitrogen atoms, making the molecule almost inert. For atmospheric N_2 to be of use to plants, it must be reduced to NH_3 by a process known as **nitrogen fixation**. All nitrogen-fixing organisms are bacteria. Some nitrogen-fixing bacteria are free-living in the soil (see Figure 37.11), whereas others live in the rhizosphere. Among this latter group, members of the genus *Rhizobium* form efficient and intimate associations with the roots of legumes (such as beans, alfalfa, and peanuts), altering the structure of the hosts' roots markedly, as will be discussed shortly.

The multistep conversion of N_2 to NH_3 by nitrogen fixation can be summarized as follows:



▼ **Figure 37.11 The roles of soil bacteria in the nitrogen nutrition of plants.** Ammonium (NH_4^+) is made available to plants by two types of soil bacteria: those that fix atmospheric N_2 (nitrogen-fixing bacteria) and those that decompose organic material (ammonifying bacteria). Although plants absorb some ammonium from the soil, they absorb mainly nitrate, which is produced from ammonium by nitrifying bacteria. Plants reduce nitrate back to ammonium before incorporating the nitrogen into organic compounds.



VISUAL SKILLS If an animal died near a root, would the plant have greater access to ammonium, nitrate, or both?

➔ **Mastering Biology Figure Walkthrough**